

How To Run It

A short guide in plain English. Follow it from top to bottom. You only do Part 1 once.

1 Install it (only the first time)

You need Python on your computer. If you can already type `python` in a terminal, you have it.

Step 1. Open a terminal in the project folder.

On Windows: open the folder `RIS PROJECT`, click the address bar at the top, type `cmd` and press Enter.

Step 2. Type this and press Enter:

```
python -m pip install -r uav_iov_simulator/requirements.txt
```

Wait two to five minutes. It downloads the libraries the app needs.

How do I know it worked?

The last line says something like `Successfully installed ...`
Warnings in yellow are normal. Ignore them.

2 Start the app

Type this and press Enter:

```
python -m streamlit run uav_iov_simulator/app.py
```

Your web browser opens by itself after about fifteen seconds.

If the browser does not open: look in the terminal. It prints a line like `Local URL : http://localhost:8501`

Copy that address into your browser. This always works.

Why `python -m` at the start?

Because plain `streamlit` is not found on this computer – Windows does not look in the folder where it was installed. Putting `python -m` in front tells Python to find it directly. It always works, so just always type it.

3 Check that everything is correct

This is optional, but it is worth doing once, and it is impressive to show.

```
python -m pytest uav_iov_simulator/tests -q
```

Wait about one minute. You want to see this at the end:

71 passed

What are these tests?

They check that the maths from the research paper actually works. The most important one checks that the paper's verification equation balances when you use real cryptography.

If that test failed, nothing else in the project would mean anything. It passes.

4 Stop the app

Click on the terminal window and press **Ctrl and C together**.

Closing the browser tab does **not** stop it. You must use Ctrl+C in the terminal.

5 Using it

When the app opens you see a grey bar on the left (the **sidebar**) and tabs across the top.

Everything starts from the sidebar. Press the buttons in this order:

1. **Initialize** builds the system
2. **Start Authentication** the drones prove who they are
3. **Distribute Group Key** they all get a shared secret

Then look at the tabs along the top to see what happened.

A button that is **grey and unclickable** is not broken. It means you have to do an earlier step first. For example you cannot give out the group key before the drones have been checked.

Two sliders worth playing with:

- **Availability** – set it to **0** and press Start Authentication. Nothing gets authenticated. This shows that the system stops completely when it loses its connection to the cloud.
- **Forged credentials** – set it to **0.10** and press Start Authentication. Everyone gets rejected, even the honest drones.

6 If something goes wrong

streamlit is not recognised	You forgot <code>python -m</code> at the start. Add it.
No such file or directory	You are in the wrong folder. You must be in <code>RIS PROJECT</code> , not inside <code>uav_iov_simulator</code> . Type <code>cd ..</code> to go up one level.
Port already in use	The app is already running in another window. Close it, or add <code>--server.port 8502</code> to the end of the command.
The page is blank	Refresh the browser once.
It looks frozen	Check the left sidebar. If the backend is set to Faithful , it is doing real cryptography and that genuinely takes a few seconds. Switch to Abstracted to make it instant.
A button does nothing	Press Reset and start again from Initialize. It takes thirty seconds.

Never try to fix things during a presentation.

Press Reset and carry on talking, or use your screenshots. A small glitch costs you nothing. Five minutes of silent typing costs you the room.

7 The two backends

In the sidebar you can choose between two settings. This confuses people, so here is the simple version.

Abstracted	Fast, instant, good for demonstrating behaviour. The maths is simplified but the protocol behaves exactly the same. Use this most of the time.
Faithful	Real cryptography on a real elliptic curve. Takes two to three seconds for eight drones. Use this once , to show that the cryptography is genuine.

The app always shows which one is active, in the sidebar and in the bar at the top of the page. Nobody can be misled about which is running.

A simplified educational implementation. Not a reproduction of the paper's benchmark environment.